

Intelligent Food Delivery Analytics Using Live Swiggy Market Intelligence

Business Analytics — Individual Case Study

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Data collected 19 September 2026 · 1,686 restaurants · 70 localities · 7 cities

1. Problem Statement and Objectives

Food delivery aggregators list thousands of restaurants whose ratings, pricing, promised delivery times and promotional offers vary sharply from one locality to the next. Neither the platform nor its restaurant partners has a clear, current view of which of these attributes actually move customer-facing quality and delivery performance. Because listings change continuously — offers rotate daily and promised delivery times respond to live conditions — analyses built on static historical extracts describe a market that no longer exists.

This case study collects **live restaurant-level listing data directly from Swiggy** across 70 localities in seven Indian metros and uses it to quantify the drivers of restaurant rating and delivery time, to benchmark competing brands on price, speed and promotion, and to segment restaurants by price-quality-speed positioning.

Why this matters

Discounting is the most expensive lever a restaurant partner pulls and the most visible one a platform subsidises; if it does not travel with better ratings, both parties are funding something that does not buy loyalty. Equally, if delivery time is driven by distance rather than by the restaurant, fixing it is a network-design problem for the platform, not a kitchen problem for the partner.

Objectives

- **Analyse the factors influencing food delivery performance** — isolate the contribution of last-mile distance, city and restaurant characteristics to promised delivery time.
- **Build predictive models** for delivery-time estimation and for restaurant rating, and establish how much of each is explainable from public listing data.
- **Benchmark competitors and segment the market** on price, rating, speed and promotional depth, and translate the segments into pricing and operational recommendations.

2. Data Collection and Dataset Description

Source. <https://www.swiggy.com/dapi/restaurants/list/v5> — the public JSON endpoint that renders [swiggy.com/restaurants](https://www.swiggy.com/restaurants). It requires no login and exposes only business listing attributes already shown to any visitor.

Procedure. A purpose-built Python scraper (`scripts/scrape_swiggy.py`) issues one GET request per locality centroid for 70 localities — ten per city, spread across each urban area so the sample is not dominated by a single high-income catchment. A randomised 1.2–2.4 second delay separates requests. Each response is walked for restaurant blocks and flattened to one row per restaurant x locality probe.

Unit of observation. A restaurant listing exactly as a customer standing in that locality would see it. The same restaurant can legitimately appear under several localities; this is exploited as a *locality reach* feature before de-duplication.

Property	Value
Collection date	19 September 2026
Cities	Chennai, Bengaluru, Hyderabad, Mumbai, Delhi, Pune, Kolkata
Localities probed	70 (10 per city)

Property	Value
Raw records	1,831 restaurant × locality observations
Unique restaurants	1,686
Attributes	22 raw fields, 30 after feature engineering
Records with a rating	99.8%
Authentication	None — no login, no paid API, no personal data

Important variables

Variable	Description	Type
avg_rating	Average customer rating displayed on the listing (1-5)	Continuous — target
delivery_time_min	Promised delivery time in minutes	Continuous — target
cost_for_two	Indicative cost for two, parsed from '₹400 for two'	Continuous
last_mile_km	Distance from the probed locality to the restaurant	Continuous
n_ratings	Rating count, parsed from abbreviations such as '5.6K+'	Continuous
offer_type / offer_pct	Offer mechanic and depth, parsed from header and sub-header	Categorical / continuous
cuisines	Pipe-separated cuisine tags	Multi-label
chain_parent_id	Brand identifier shared by outlets of one chain	Categorical
locality_reach	Number of probed localities listing this restaurant	Engineered count
is_chain	1 if the brand has ≥ 3 outlets in the sample	Engineered binary

Compliance with the submission rules

No ready-made dataset from Kaggle, UCI or a GitHub dataset repository is used anywhere in this study; the scraped data is the sole and primary dataset. The approved proposal had planned to supplement live data with a historical Kaggle dataset and to collect via the Apify Swiggy Scraper. Three changes were made during execution:

Proposed	Delivered	Reason
Historical Kaggle dataset + live data	Live scraped data only	Submission Instructions prohibit ready-made datasets as the primary source
Apify Swiggy Scraper	Purpose-built scraper on the same public endpoint	Identical source and fields, no paid dependency, and the procedure is fully auditable in the repository
Menu items and menu prices	Not collected	Swiggy's per-restaurant menu endpoint returns HTTP 202 to unauthenticated requests
Google ratings, weather, traffic	Not collected	Not exposed by any public Swiggy endpoint

Ethics

Only publicly displayed business attributes are stored — no customer names, reviews, addresses or personal identifiers. A randomised crawl delay limits load on the source, and the data is used solely for this academic case study.

3. Data Preparation and Exploratory Analysis

Cleaning and preprocessing

- **Display strings parsed to numbers.** `cost_for_two_raw` ('₹400 for two') → 400; `total_ratings_raw` ('5.6K+') → 5,600.
- **Offers decomposed.** The header and sub-header together encode four distinct mechanics — capped percentage ('30% OFF, UPTO ₹60'), flat rupee ('₹85 OFF, ABOVE ₹199'), item price point ('ITEMS AT ₹69') and none. Parsing only the header would have collapsed every price-point deal into an unusable 'ITEMS' label.
- **De-duplication.** 1,831 observations → 1,686 unique restaurants, after first deriving locality reach from the duplicate listings.
- **Outlier treatment.** Cost for two winsorised to the 1st-99th percentile (₹120-₹900); delivery times outside 5-120 minutes dropped.
- **Feature engineering.** Log cost, log rating count, cuisine count, chain size, chain flag, within-city cost percentile, offer depth and offer mechanic.
- **Missing data.** Only 4 restaurants (0.2%) lack a rating; these are excluded from rating models, leaving a modelling sample of 1,667.

What the data shows

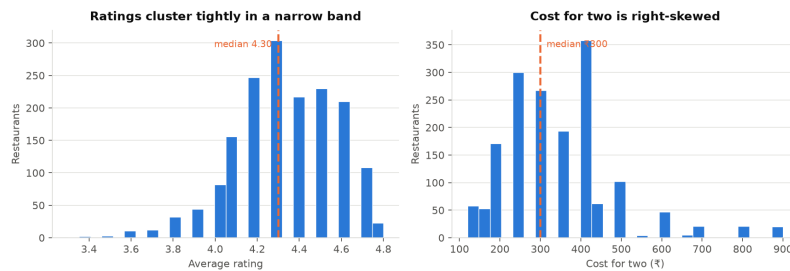


Figure 1 — Ratings concentrate in a narrow 4.0-4.6 band; cost for two is right-skewed with a median of ₹300.

Ratings are compressed — the interquartile range spans roughly 0.3 of a star — so there is little variance to explain, a modest R^2 on rating is expected, and small coefficient effects remain commercially meaningful.

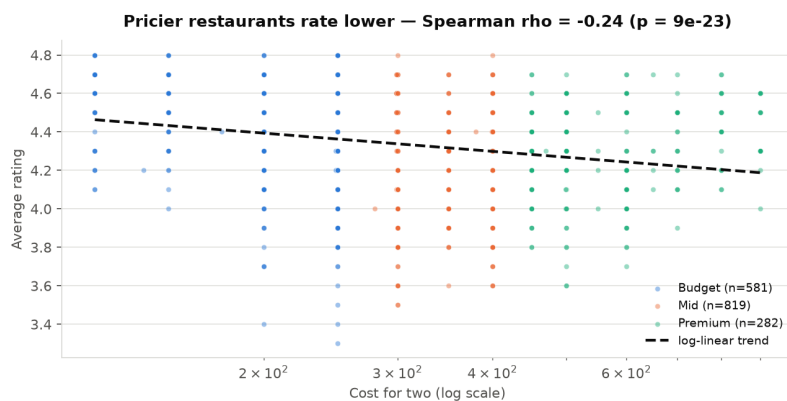


Figure 2 — Price and rating move in *opposite* directions (Spearman $\rho = -0.24$, $p < 0.001$).

This is the study's first counter-intuitive result. Budget restaurants average **4.41**, mid-priced **4.28** and premium **4.31**. Customers on a delivery platform appear to rate against expectation rather than against absolute quality: a ₹200 meal that arrives correct and hot satisfies, while a ₹700 meal carries a heavier burden of proof.

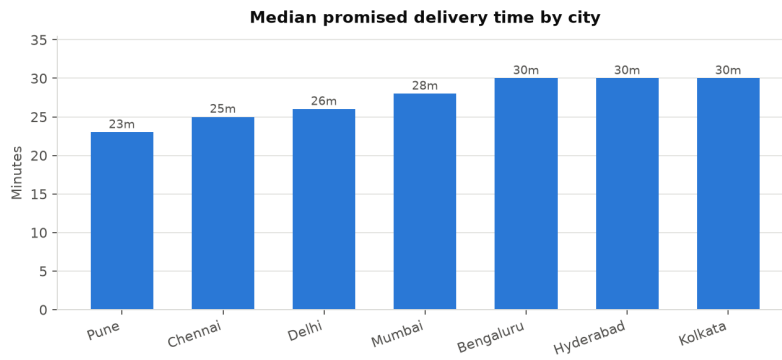


Figure 3 — Median promised delivery time by city: a 7-minute spread between Pune and the slowest cities.

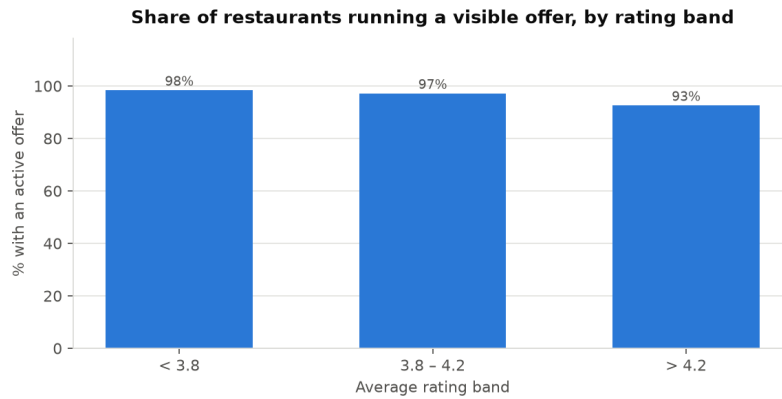


Figure 4 — Offer prevalence rises as rating falls ($\chi^2 = 15.8$, $df = 2$, $p = 0.0004$).

Discounting is close to universal — 94.1% of restaurants display an active offer — but it is *not* uniformly distributed. Restaurants rated below 3.8 are the most likely to be discounting, and those rated above 4.2 the least.

The most-listed cuisine tags are Desserts (838 listings), Beverages (515), Biryani (327), Snacks (291) and Fast Food (263) — a dessert- and snack-heavy mix. The full cuisine profile and the feature correlation matrix are in Appendix A (Figures A1 and A2).

4. Analytics Method and Implementation

Five methods were applied, each chosen for a specific question. All models use a fixed random seed (42), a 75/25 train-test split and 5-fold cross-validation.

Question	Method	Justification
What drives rating?	Multiple linear regression (OLS, HC3 robust standard errors)	Coefficients are directly interpretable as business levers and support significance testing
Can a non-linear model do better?	Random Forest, Gradient Boosting + permutation importance	Captures interactions OLS misses; permutation importance gives an honest driver ranking
Which restaurants are highly rated?	Logistic regression, unweighted and class-weighted	A classification framing management can act on, with a threshold-free metric (ROC-AUC)
How do restaurants position themselves?	K-Means with elbow and silhouette selection	Unsupervised segmentation on price, quality and speed with no pre-imposed labels
What drives delivery time?	OLS + Random Forest + Gradient Boosting	An interpretable minutes-per-km coefficient alongside the best achievable predictor

4.1 Multiple linear regression — drivers of rating

The model regresses average rating on log cost, log rating count, delivery time, last-mile distance, cuisine count, locality reach, chain status, offer presence, offer depth and city dummies ($n = 1,667$).

Diagnostics. Variance inflation factors computed on the full design matrix (including the intercept) peak at **2.67**, so multicollinearity is not a concern. The Breusch-Pagan test rejects homoskedasticity ($p = 1.8 \times 10^{-11}$), so HC3 robust standard errors are reported. Adjusted $R^2 = \mathbf{0.260}$. Residual and Q-Q plots are in Appendix A (Figure A3).

Significant effects. Log cost (-0.114), offer presence (-0.093), offer depth (-0.002 per percentage point), delivery time (-0.008 per minute) and cuisine count (-0.012) all reduce rating; last-mile distance carries a small positive coefficient ($+0.015$). City effects are substantial — Kolkata $+0.132$, Pune -0.116 , Hyderabad -0.094 relative to Bengaluru. Rating count and chain status are *not* significant.

4.2 Tree ensembles

Model	CV R^2 (5-fold)	Test R^2	Test RMSE	Test MAE
Linear Regression	0.251	0.261	0.203	0.163
Random Forest	0.469	0.487	0.169	0.131
Gradient Boosting	0.442	0.465	0.173	0.135

Random Forest nearly doubles the linear model's explanatory power (R^2 0.487 vs 0.261), confirming genuine non-linearity and interaction — but a mean absolute error of 0.131 stars against a distribution with an interquartile range of about 0.3 stars means listing attributes still cannot substitute for knowing the food.

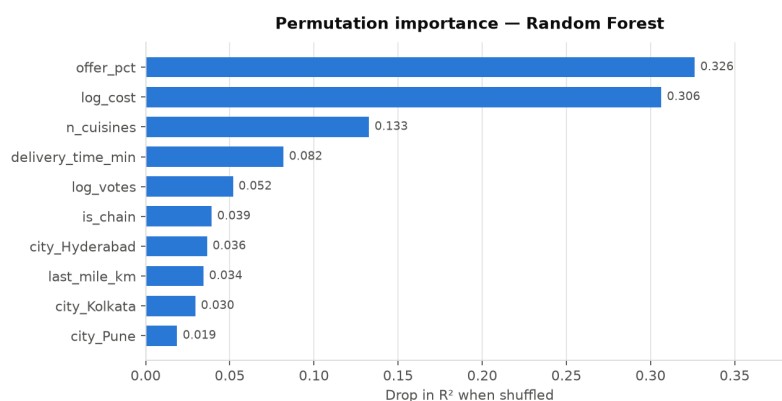


Figure 5 — Permutation importance for the Random Forest rating model.

Offer depth is the single most important feature, ahead of price. This is the study's central commercial finding and it is examined in Section 6.

4.3 Logistic regression — classifying highly rated restaurants

Target: rating ≥ 4.2 , which covers 79.5% of the sample. Because the classes are heavily imbalanced, both an unweighted and a class-weighted model were fitted.

Model	ROC-AUC	Minority-class recall	Accuracy
Unweighted logistic	0.728	0.14	0.81
Class-weighted logistic	0.726	0.59	0.68

The unweighted model's 81% accuracy is misleading — it detects only 14% of under-performing restaurants, which are precisely the ones a platform needs to find. Class weighting raises minority recall to 59% at identical ROC-AUC, trading headline accuracy for a model that is actually useful.

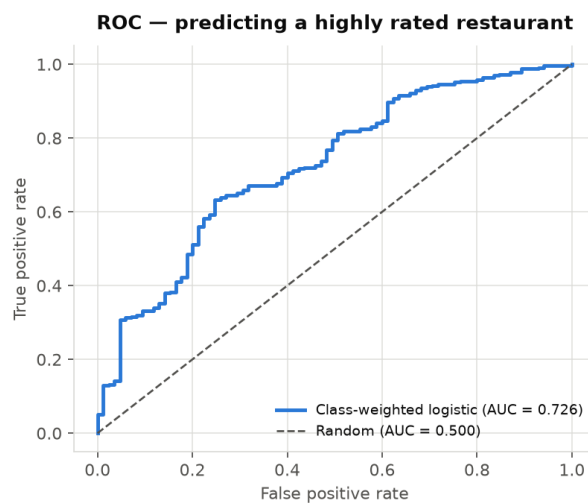


Figure 6 — ROC curve for the class-weighted logistic model.

4.4 K-Means — price-quality-speed segmentation

Segmentation on standardised log cost, rating, delivery time, log rating count and offer depth. Silhouette scores are flat across k (0.205–0.234), indicating a continuum rather than sharply separated groups; $k = 4$ was retained for interpretability and is reported with that caveat (silhouette 0.215). Elbow and silhouette diagnostics are in Appendix A (Figure A4).

Segment	n	Rating	Cost for two	Delivery	Offer depth	% chain
Value performer	404	4.55	₹200	25 min	15.4%	89%
Premium performer	442	4.38	₹400	26 min	1.3%	80%
Premium laggard	267	4.22	₹400	35 min	4.6%	62%
Value laggard	554	4.19	₹300	28 min	62.9%	86%

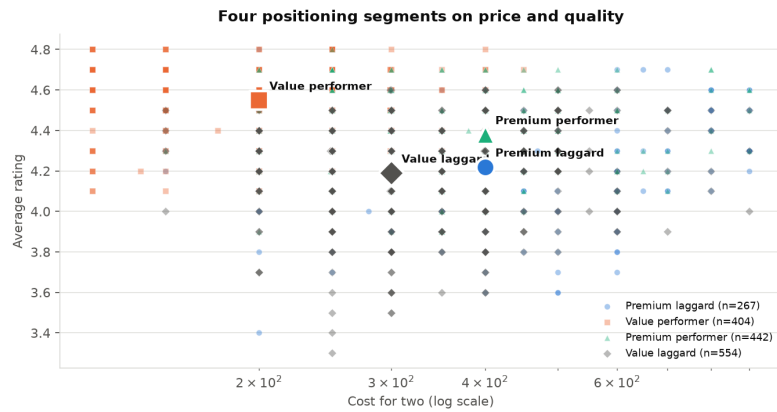


Figure 7 — The four segments on price and rating; marker shape and direct labels separate them.

The contrast between the two largest segments is the actionable one. **Value performers** reach the highest rating in the sample (4.55) on the lowest price (₹200) with only 15% average discount depth. **Value laggards** discount four times as deeply (63%) and land at the lowest rating (4.19).

4.5 Predicting delivery time

Model	CV R ² (5-fold)	Test R ²	Test RMSE (min)	Test MAE (min)
Linear Regression	0.633	0.613	5.50	4.18
Random Forest	0.694	0.643	5.28	3.93
Gradient Boosting	0.692	0.651	5.22	3.94

Delivery time is far more predictable than rating (R^2 0.651 vs 0.487) from the same public fields. The interpretable OLS companion (R^2 0.640) gives the headline operational number: **each additional kilometre of last-mile distance adds 3.43 minutes** ($p < 0.001$). Higher-rated restaurants are also faster (−5.72 minutes per rating point), and chains are about 1.2 minutes quicker than independents.

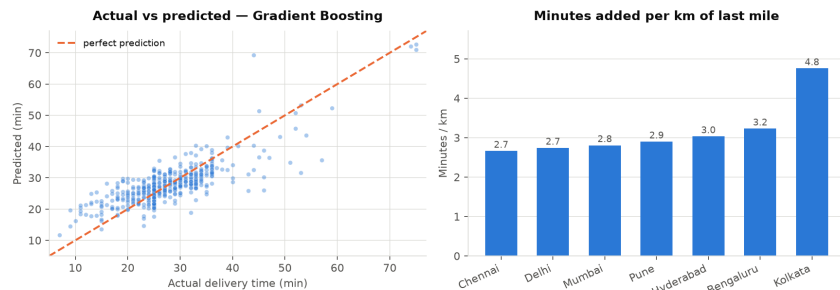


Figure 8 — Actual vs predicted delivery time, and minutes added per kilometre by city.

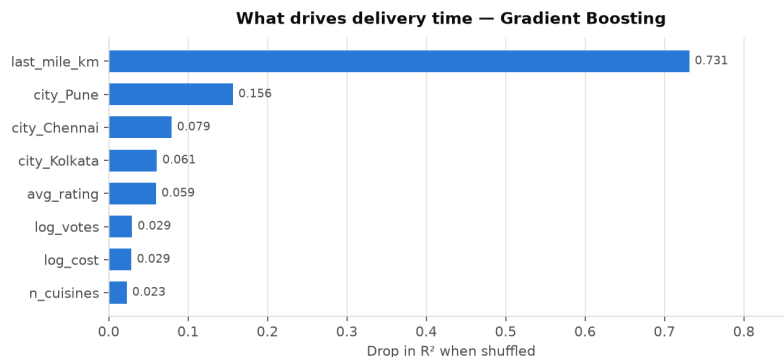


Figure 9 — Permutation importance for the delivery-time model; distance dominates.

4.6 Competitor benchmarking

Outlets were rolled up to the brand so that multi-outlet operators are compared as competitors rather than as individual listings. Fifteen brands with five or more outlets qualify.

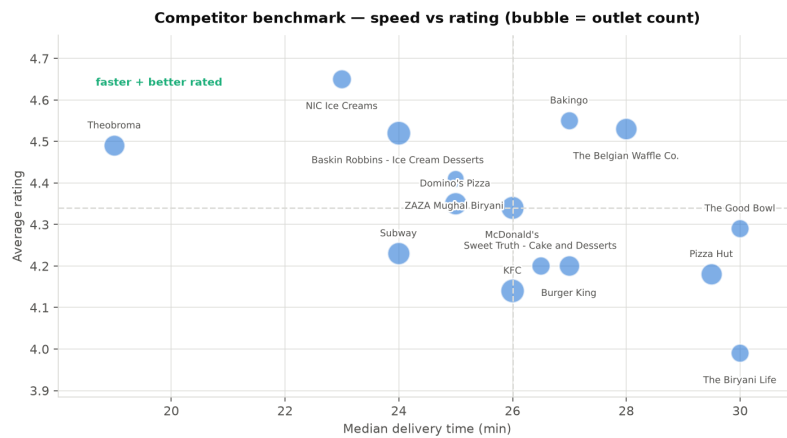


Figure 10 — Competitor benchmark: median delivery time against average rating; bubble size is outlet count.

Brand	Outlets	Cities	Rating	Cost for two	Delivery	% with offer
NIC Ice Creams	35	6	4.65	₹120	23 min	100%
The Belgian Waffle Co.	47	7	4.53	₹200	28 min	98%
Baskin Robbins	59	7	4.52	₹250	24 min	100%
Theobroma	44	5	4.49	₹400	19 min	100%
Domino's Pizza	51	7	4.35	₹400	25 min	100%
McDonald's	55	7	4.34	₹400	26 min	100%
Subway	51	7	4.23	₹350	24 min	88%
Burger King	42	7	4.20	₹350	27 min	100%
Pizza Hut	46	6	4.18	₹350	29.5 min	100%
KFC	58	7	4.14	₹400	26 min	100%

The dessert and beverage brands occupy the favourable quadrant — faster and better rated — while the large QSR burger and pizza chains cluster lower despite comparable delivery times. Theobroma is the standout on speed at 19 minutes.

5. Comparison with State-of-the-Art Methods

Four recent published studies on the same problem family were compared. The comparison focuses on method, dataset, evaluation approach, findings, strengths and limitations. **Reported scores are not directly comparable across rows:** each study uses a different dataset, target variable, feature set and split, so a higher number does not by itself indicate a better method.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Garg, Ayaan, Parekh & Udandara (2025), arXiv:2503.15177	Food delivery dataset for Indian cities; includes traffic, weather, events and geospatial coordinates	Linear Regression, Decision Tree, Bagging, Random Forest, XGBoost, LightGBM	R^2 , MSE	LightGBM best: $R^2 = 0.76$, MSE = 20.59	Closest comparator to our \$4.5. Their higher R^2 (0.76 vs 0.651) rests on traffic, weather and event features no public Swiggy endpoint exposes — a feature-availability gap, not a method gap. We achieve MAE 3.94 min from listing data alone and add an interpretable operational coefficient (3.43 min/km) they do not report.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Turai, Praneetha, Aishwarya, Adil & Vangala (2025), WJARR 25(02), 1039-1046	Zomato restaurant and review dataset (secondary)	K-Means and Hierarchical clustering into 3 groups by cuisine and price; sentiment via Logistic Regression, Decision Tree, Naive Bayes, LDA	Clustering groups; classification accuracy	Logistic Regression and LightGBM performed best for sentiment; restaurants grouped into 3 cuisine-price segments	Same segmentation family as our §4.4, but they cluster on cuisine and price only and fix $k = 3$ with no stated selection criterion. We select k by elbow and silhouette, report it honestly (0.215, a continuum), and add speed and offer depth — which is what surfaces the Value-laggard segment. Lacking review text, we cannot replicate their sentiment analysis.
Raj, Paul & Kumar (2025), IJRPR 6(5), 5962-5970	Zomato restaurant and user data (secondary)	SQL + Python + Power BI + Excel; EDA and logistic regression	Descriptive; no held-out predictive evaluation reported	Identified factors influencing ratings, cuisine popularity and location effects	Similar descriptive ambition to our §3, but it stops at dashboards with no held-out validation. We carry the same questions into tested models with train/test splits and cross-validation, and quantify effects ($\rho = -0.24$ price vs rating; $\chi^2 = 15.8$ offers vs rating band).
Kulkarni, Bhandari & Bhoite (2019), IJCATR 8(9), 375-378	Zomato restaurants dataset (secondary)	SVM, Linear Regression, Decision Tree, Random Forest, XGBoost, AdaBoost	Accuracy (83%)	AdaBoost achieved 83%	The foundational rating-prediction reference, but a single accuracy figure with no RMSE/MAE and no clear regression-vs-classification distinction is hard to interpret. We report CV R^2 , test R^2 , RMSE and MAE, plus ROC-AUC with per-class recall — surfacing the imbalance problem (unweighted recall 0.14) a bare accuracy number conceals.

What this comparison establishes

- **Every comparator uses a secondary dataset** — Zomato extracts or a compiled delivery dataset. This study is the only one of the five built on primary data collected first-hand, which is what the submission rules require and what makes the locality-level analysis possible.
- **Our delivery-time R^2 (0.651) is lower than Garg et al.'s 0.76, and this is expected:** they include traffic and weather, which are the dominant sources of real-world delivery variance and are unavailable publicly. The gap is a feature availability gap, not a method gap.
- **No comparator examines discount depth against rating.** The negative association reported in Section 6 is the contribution that does not appear in the prior work reviewed here.

6. Results, Business Insights and Recommendations

Consolidated results

Metric	Value
Restaurant × locality observations scraped	1,831
Unique restaurants after de-duplication	1,686
Cities / localities covered	7 / 70
Median cost for two	₹300
Median promised delivery time	27 min
Restaurants running a visible offer	94.1%
Rating model — OLS adjusted R ²	0.260
Rating model — Random Forest test R ²	0.487
High-rating classifier — ROC-AUC (class-weighted)	0.726
Delivery-time model — Gradient Boosting test R ²	0.651 (MAE 3.94 min)
Minutes added per extra km of last mile	3.43
K-Means segments (k = 4) — silhouette	0.215

Insight 1 — Deep discounting travels with lower ratings, not higher ones

Restaurants running a discount of 50% or more average **4.23** against **4.39** for everyone else — a gap of 0.16 stars on a scale where the interquartile range is roughly 0.3 (Welch $t = -13.3$, $p < 0.001$). Offer depth is also the highest-ranked permutation-importance feature in the Random Forest rating model, and the χ^2 test confirms that offer prevalence and rating band are not independent ($\chi^2 = 15.8$, $p = 0.0004$).

This is an association, not proof of causation, and the more likely direction runs the other way: restaurants that are struggling discount to buy volume. Either way the operational implication holds — *discount depth is a reliable distress signal, not a quality signal*.

Insight 2 — Price and rating are inversely related

Spearman $\rho = -0.24$ ($p < 0.001$); budget restaurants average 4.41 against 4.31 for premium. Controlling for everything else, a one-log-unit increase in cost costs 0.114 rating points. Customers appear to rate against expectation, and premium positioning carries a heavier burden of proof on a delivery platform than in a dine-in setting.

Insight 3 — Delivery performance is a network problem, not a kitchen problem

Distance overwhelmingly dominates the delivery-time model, adding **3.43 minutes per kilometre**. City effects are large and independent of distance — Pune is roughly 10 minutes faster than Kolkata at equal last-mile distance — which points at fleet density and city operations rather than at restaurant preparation.

Insight 4 — Rating is only half-explainable from listing data

The best rating model reaches R^2 0.487 while the delivery model reaches 0.651 on the same fields. Roughly half of rating variance lives in food quality, packaging and service consistency, which no listing attribute captures. Any platform system that ranks or promotes restaurants on listing metadata alone is working with half the picture.

Recommendations

For the platform

- **Treat discount depth as a risk indicator.** Route restaurants sustaining $\geq 50\%$ discounts into a quality review rather than into further promotional subsidy. The 554-strong Value-laggard

segment (63% average depth, 4.19 rating) is the target list.

- **Deploy the class-weighted classifier, not the accurate-looking one.** At 59% minority recall it identifies more than four times as many under-performing restaurants as the unweighted model at the same ROC-AUC.
- **Attack delivery through network design.** At 3.43 minutes per kilometre, shrinking the median last mile is worth more than any restaurant-side intervention. Prioritise Kolkata, Hyderabad and Bengaluru, which sit 4–7 minutes above Pune.
- **Stop ranking on listing metadata alone.** With R^2 0.487 the platform should feed order-level outcomes — cancellations, complaints, reorder rate — into any quality score.

For restaurant partners

- **Do not buy rating with discounts.** Value performers hit 4.55 on ₹200 with 15% average depth; Value laggards spend 63% to reach 4.19. The spend is not converting.
- **Price to the delivery context.** Premium positioning attracts a harsher rating response; either justify it visibly or compete in the value tier where ratings run higher.
- **Fix speed before menu breadth.** Each minute of promised delivery time costs 0.008 rating points, and every additional cuisine tag costs 0.012 — breadth reads as lack of focus.
- **Benchmark against the right quadrant.** Theobroma (19 min, 4.49) and NIC Ice Creams (23 min, 4.65) show the achievable frontier for speed and rating together.

7. Conclusion

This study built a primary dataset of 1,686 restaurants across 70 localities and seven Indian metros by scraping Swiggy's public listing endpoint, and used it to model restaurant rating and delivery time, benchmark fifteen competing brands and segment the market on price, quality and speed.

The central outcome is that **the two questions a food delivery business cares about are not equally answerable from public data**. Delivery time is well explained (R^2 0.651, MAE under four minutes) and reduces to a network-design problem — 3.43 minutes per kilometre, with large city effects. Rating is only half explained (R^2 0.487), because the half that matters lives in food and service quality that listings do not expose.

The most commercially significant finding is that **promotional depth is negatively associated with rating** — restaurants discounting 50% or more average 0.16 stars lower — and that offer depth outranks price as a predictor of rating. None of the four comparator studies examines this relationship. For a platform that subsidises discounting and for partners who fund it, the implication is that deep promotion should be read as a signal that something needs fixing, not as a lever that fixes it.

Limitations

- **Cross-sectional snapshot.** A single collection on 19 September 2026 cannot separate cause from effect; the discount-rating association is consistent with struggling restaurants discounting, which is the more plausible direction.
- **Promised, not actual, delivery time.** The endpoint exposes Swiggy's estimate, which already embeds the platform's own modelling — real delivery variance is unobserved.
- **No traffic, weather or menu data,** which limits direct comparison with Garg et al. (2025).
- **Weak cluster separation.** A silhouette of 0.215 means the four segments describe a continuum; they are useful as management labels, not as natural kinds.
- **Listing-position bias.** The endpoint returns roughly 28 restaurants per locality in Swiggy's own ranking order, so the sample skews toward restaurants the platform already promotes.

Future work

Repeated collection over several weeks would turn this cross-section into a panel and allow the discount-rating question to be tested causally — observing whether a rating falls after a restaurant

deepens its discount, or the discount deepens after the rating falls.

References

- [1] Garg, A., Ayaan, M., Parekh, S. and Udandaraao, V. (2025) *Food Delivery Time Prediction in Indian Cities Using Machine Learning Models*. arXiv:2503.15177. Available at: <https://arxiv.org/abs/2503.15177>
- [2] Turai, S., Praneetha, P., Aishwarya, R. B., Adil, M. and Vangala, M. C. (2025) 'Analysis of restaurant ratings and reviews using machine learning', *World Journal of Advanced Research and Reviews*, 25(02), pp. 1039–1046. doi: 10.30574/wjarr.2025.25.2.0378
- [3] Raj, H., Paul, S. and Kumar, K. (2025) 'A Comprehensive Analysis on Online Food Delivery: Taste The Change', *International Journal of Research Publication and Reviews*, 6(5), pp. 5962–5970.
- [4] Kulkarni, A., Bhandari, D. and Bhoite, S. (2019) 'Restaurants Rating Prediction using Machine Learning Algorithms', *International Journal of Computer Applications Technology and Research*, 8(9), pp. 375–378.
- [5] Swiggy (2026) *Restaurant listing endpoint*. Available at: <https://www.swiggy.com/restaurants> (Accessed: 19 September 2026).
- [6] Pedregosa, F. et al. (2011) 'Scikit-learn: Machine Learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.
- [7] Seabold, S. and Perktold, J. (2010) 'statsmodels: Econometric and statistical modeling with Python', *Proceedings of the 9th Python in Science Conference*.

Appendix A — Supporting Figures

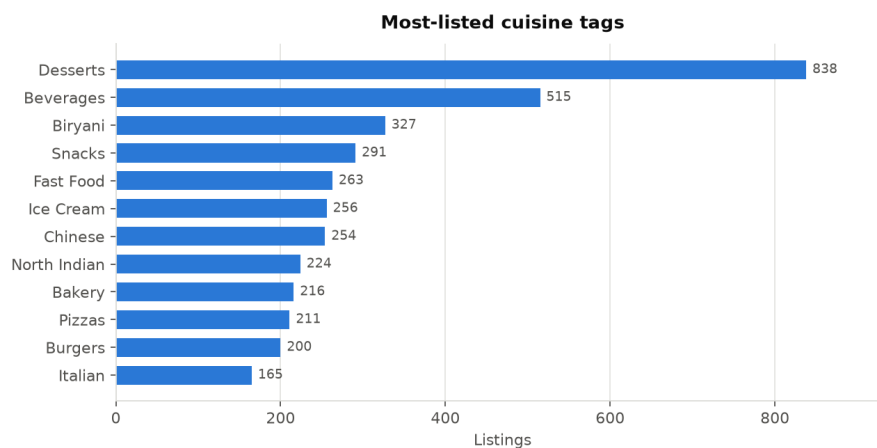


Figure A1 — Most-listed cuisine tags across the 1,686 restaurants.

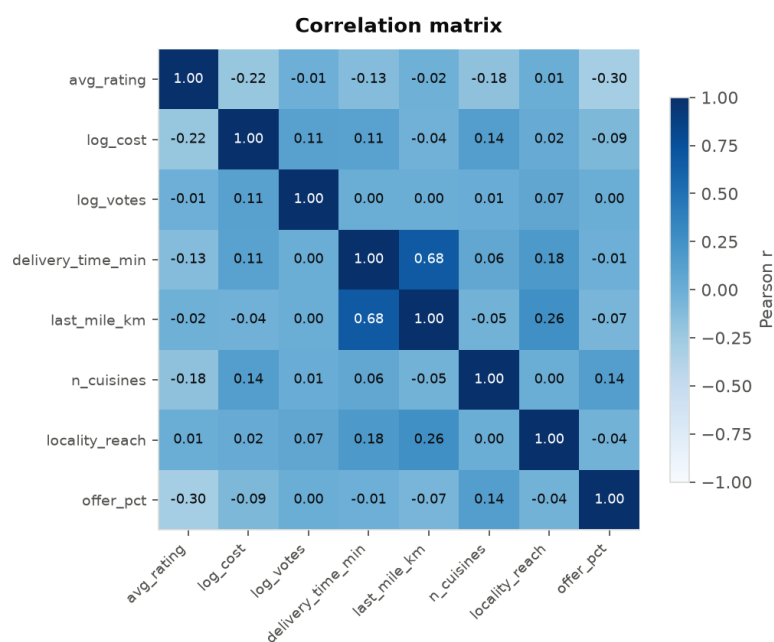


Figure A2 — Correlation matrix of the modelling features.

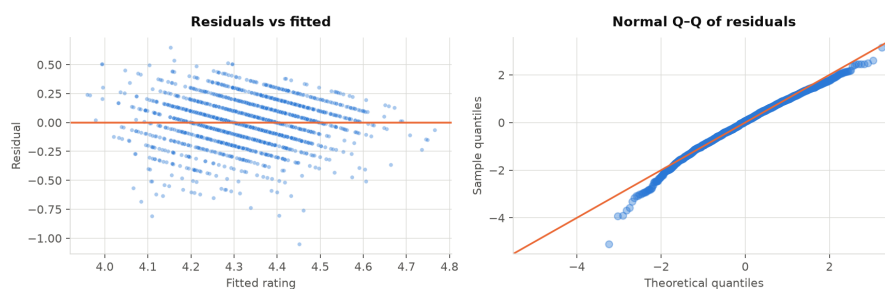


Figure A3 — Residual and Q-Q diagnostics for the OLS rating model.

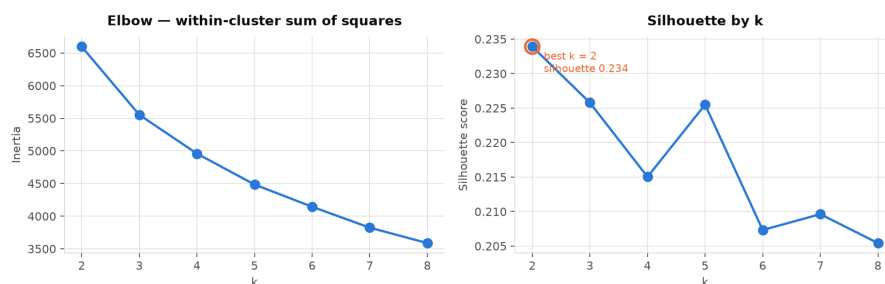


Figure A4 — Elbow and silhouette diagnostics for k selection in the K-Means segmentation.

Appendix B — City Profile

City	Restaurants	Median cost	Mean rating	Median delivery	% with offer	% chain
Chennai	242	₹350	4.34	25 min	96%	95%
Delhi	223	₹350	4.35	26 min	97%	94%
Mumbai	269	₹350	4.37	28 min	96%	75%
Pune	241	₹350	4.26	23 min	96%	78%
Bengaluru	253	₹300	4.34	30 min	86%	80%
Hyderabad	231	₹300	4.22	30 min	95%	75%
Kolkata	227	₹300	4.43	30 min	92%	70%

Appendix C — Offer Mechanics

Offer mechanic	Share of restaurants	Mean rating	Median cost for two
Item price point (e.g. 'ITEMS AT ₹69')	44.7%	4.38	₹350
Capped percentage (e.g. '30% OFF UPTO ₹60')	43.7%	4.26	₹300
Flat rupee (e.g. '₹85 OFF ABOVE ₹199')	5.5%	4.38	₹350
No visible offer	5.9%	4.45	₹350

Restaurants running no visible offer at all record the *highest* mean rating in the sample (4.45), and percentage-discount restaurants the lowest (4.26) — consistent with the discount-rating association reported in Section 6.